

RAG Threads Application

Flask + Neon + Chroma Cloud +
Backblaze B2 + OpenAI

Agenda

- Problem & goal
- Architecture overview
- Core components
- Data flow
- Storage & vector search
- Chat + RAG pipeline
- Admin & security
- Evaluation metrics
- Testing & learnings

Goal

- Multi-tenant RAG app for PDF Q&A
- Global and user-private documents
- No general chat without a document
- Admin management and audit logs

Tech Stack

- Backend: Flask + SQLAlchemy + Alembic
- DB: Neon Postgres
- Vector DB: Chroma Cloud
- Storage: Backblaze B2 (S3 compatible)
- LLM: OpenAI (gpt-4o)
- Frontend: Server-rendered templates + JS

Configuration Highlights

- MAX_UPLOAD_MB = 100
- CHUNK_SIZE = 1000
- CHUNK_OVERLAP = 200
- TOP_K_CHUNKS = 5
- MAX_COMPLETION_TOKENS = 4000
- RAG_TEMPERATURE = 0.7

Project Structure

- app/models: ORM models
- app/routes: API + views
- app/services: auth, storage, RAG, evaluation
- app/templates + static: UI
- migrations: Alembic
- tests: unit & integration

app/models Summary

- `user.py`: accounts + roles
- `document.py`: PDF metadata + scope
- `thread.py`: conversation metadata
- `message.py`: user/assistant content
- `message_evaluation.py`: LLM-judge scores
- `ingestion_job.py`: processing queue
- `activity_log.py`: audit trail

app/routes Summary

- auth.py: register/login/OAuth
- documents.py: upload/presign/confirm
- threads.py: CRUD threads
- chat.py: RAG responses + metrics
- admin.py: admin-only management
- views.py: HTML pages

app/services Summary

- auth_service: sessions + guards
- storage_service: B2 presigned URLs
- vector_service: Chroma search/upsert
- rag_service: retrieval + prompt
- ingestion_service: PDF processing
- evaluation_service: LLM judge

Key Models

- users, documents, threads, messages
- ingestion_jobs for async processing
- activity_logs for audit
- message_evaluations for LLM-judge metrics

Upload Flow

- Client requests presigned URL
- Direct upload to B2
- Confirm upload creates document + ingestion job
- Worker extracts text, chunks, embeds

Ingestion Worker

- Poll queued ingestion jobs
- Download PDF from B2
- Extract text per page
- Chunk with overlap
- Embed with OpenAI
- Upsert to Chroma
- Update job status

Ingestion & Chunking

- Download PDF from B2
- Extract per-page text
- Chunk with overlap
- Generate embeddings
- Upsert into Chroma with metadata

Chroma Cloud Usage

- Single collection for vectors
- Metadata filters: document_id, scope, owner_user_id
- Search returns top-k chunks

Backblaze B2 Usage

- Store original PDFs
- Store extracted images (optional)
- Presigned upload/download
- Delete by prefix for cleanup

RAG Chat Pipeline

- Retrieve top-k chunks from Chroma
- Assemble prompt with excerpts
- Call OpenAI for answer
- Store response + citations

Prompting & Temperature

- System prompt enforces grounded answers
- RAG_TEMPERATURE = 0.7
- Max completion tokens = 4000
- Message history included for context

End-to-End Flow

Upload PDF

Presign + B2
Upload

Confirm + DB Row

Ingest + Chunk

Chroma Upsert

Chat + Retrieve

LLM Answer

LLM Judge

Authorization

- login_required, admin_required
- owns_document, owns_thread
- Admin-only management endpoints

Admin Panel

- User management
- Document management
- Activity log
- Re-ingest / delete chunks

Deletion Semantics

- Soft-delete records
- Delete vectors by document_id
- Delete B2 objects by prefix
- Cascade on user deletion

Evaluation (LLM Judge)

- Faithfulness (0–1)
- Citation precision (0–1)
- Groundedness (0–1)
- Scores stored per message

Evaluation Workflow

- Extract citations from answer text
- Judge compares answer to excerpts
- Heuristics penalize missing citations
- Scores shown to users

Evaluation UX

- Metrics panel shows scores (0–100)
- Heuristics penalize missing citations
- Off-topic answers drop scores

Testing

- Pytest unit tests for models/auth
- Integration tests for access rules
- Evaluation persistence test

Security Considerations

- Role-based access control
- Scope filtering in vector search
- No cross-tenant data leakage

Deletion Flow

- Soft-delete records in Postgres
- Delete vectors by document_id
- Delete B2 objects by prefix
- Admin cascade delete users

Performance Notes

- Batch upserts to Chroma
- Streaming chat for UX
- Background ingestion worker

Key Points

- Why presigned uploads
- How metadata filters enforce access
- LLM judge for quality signals
- Operational tradeoffs

Future Improvements

- Better citation extraction
- Asynchronous judge jobs
- More evaluation metrics
- Team/tenant support

Thank You

- Questions?
- Demo ready